

Causal inference and missing data

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Causal inference is a missing data problem

- Fundamental problem of causal inference (Holland, 1986)
 - for each unit at most one of the potential outcomes is observed
 - estimating causal effects requires properly handling the missing potential outcome
- Four variables are associated with each unit: $\{Y_i(0), Y_i(1), Z_i, \mathbf{X}_i\}$
 - $Y_i^{\text{obs}} = Y_i(Z_i)$ is observed; $Y^{\text{mis}} = Y_i(1 - Z_i)$ is missing
 - $Y_i^{\text{obs}} = Y_i(1)Z_i + Y_i(0)(1 - Z_i)$; $Y^{\text{mis}} = Y_i(1)(1 - Z_i) + Y_i(0)Z_i$
- In principal stratification problems, $U_i = (S_i(1), S_i(0))$ is missing for each unit

Missing data

- Full data O_i : the vector containing all the relevant data, observed or unobserved
- Missing data indicator R_i : same dimension as O_i
- $\mathbf{O} = (O_1, O_2, \dots, O_n)'$ and $\mathbf{R} = (R_1, R_2, \dots, R_n)'$
- $\text{pr}(\mathbf{R}, \mathbf{O}) = \text{pr}(\mathbf{O})\text{pr}(\mathbf{R} \mid \mathbf{O})$
 - parameters associated with $\mathbf{O}$: θ
 - parameters associated with $\mathbf{R}$: ϕ
 - θ and ϕ are distinct
- Missing data mechanism
 - missing completely at random (**MCAR**): $\text{pr}(\mathbf{R} \mid \mathbf{O}, \theta, \phi) = \text{pr}(\mathbf{R} \mid \phi)$
 - missing at random (**MAR**): $\text{pr}(\mathbf{R} \mid \mathbf{O}, \theta, \phi) = \text{pr}(\mathbf{R} \mid \mathbf{O}^{\text{obs}}, \phi)$
 - missing not at random (**MNAR**): $\text{pr}(\mathbf{R} \mid \mathbf{O}, \theta, \phi) = \text{pr}(\mathbf{R} \mid \mathbf{O}^{\text{obs}}, \mathbf{O}^{\text{mis}}, \phi)$

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- Scenario 3: all variables have missing values (R_1, R_2, R_3)
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- More detailed discussion in Mealli and Rubin (2015)

A missing data perspective of causal inference

- In a causal setting
 - $O_i = \{Y_i(1), Y_i(0), Z_i, \mathbf{X}_i\}$
 - $R_i = (Z_i, 1 - Z_i, 1, \mathbf{1}_p)$
 - $O_i^{\text{obs}} = \{Y_i^{\text{obs}}, Z_i, \mathbf{X}_i\}$, $O_i^{\text{mis}} = Y_i^{\text{mis}} = Y_i(1 - Z_i)$
 - Causal inference is a missing data problem with a particular missing pattern
- Treatment assignment mechanism is a special case of missing data mechanism
 - Randomization: $\text{pr}\{Z_i \mid Y_i(1), Y_i(0), \mathbf{X}_i, \theta, \phi\} = \text{pr}(Z_i \mid \phi) \rightsquigarrow \text{MCAR}$
 - Non-ignorability:
 $\text{pr}\{Z_i \mid Y_i(1), Y_i(0), \mathbf{X}_i, \theta, \phi\} = \text{pr}(Z_i \mid Y_i^{\text{obs}}, Y_i^{\text{mis}}, \mathbf{X}_i, \phi) \rightsquigarrow \text{MNAR}$
 - Ignorability: $\text{pr}\{Z_i \mid Y_i(1), Y_i(0), \mathbf{X}_i, \theta, \phi\} = \text{pr}(Z_i \mid \mathbf{X}_i, \phi) \rightsquigarrow \text{MAR}$
 - stratified experiments, observational studies without latent confounding
 - sequential experiments, sequentially ignorable assignment mechanism

Why consider MAR?

- Complete case analysis is valid under MCAR
 - only consider units with $R_i = 1$
 - too ideal to be true
- MNAR: no assumption on $R_i \rightsquigarrow$ inference is impossible
- MAR: a realistic and practical scenario in between
 - many classic methods rely on MAR

Likelihood inference with missing data

- MLE for θ and ϕ
- Likelihood for the observed data

$$\begin{aligned} & L(\theta, \phi \mid \mathbf{R}, \mathbf{O}^{\text{obs}}) \\ = & \text{pr}(\mathbf{R}, \mathbf{O}^{\text{obs}}; \theta, \phi) \\ = & \int \text{pr}(\mathbf{R}, \mathbf{O}^{\text{obs}}, \mathbf{O}^{\text{mis}}; \theta, \phi) d\mathbf{O}^{\text{mis}} \\ = & \int \text{pr}(\mathbf{R} \mid \mathbf{O}^{\text{obs}}, \mathbf{O}^{\text{mis}}; \theta, \phi) \text{pr}(\mathbf{O}^{\text{obs}}, \mathbf{O}^{\text{mis}}; \theta) d\mathbf{O}^{\text{mis}} \end{aligned}$$

- Without any assumptions, MLE requires considering missing data mechanism $\text{pr}(\mathbf{R} \mid \mathbf{O}^{\text{obs}}, \mathbf{O}^{\text{mis}}; \theta, \phi)$

Likelihood inference under MAR

- Likelihood for the observed data

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- We can ignore the missing data mechanism under MAR
 - MLE for θ : $\text{argmax}_{\theta} L(\theta \mid \mathbf{O}^{\text{obs}})$

Expectation-maximization (EM) algorithm (Dempster, Laird and Rubin, 1977)

- MLE with missing data under MAR $\rightsquigarrow$ maximize

$$L(\theta; Y^{\text{obs}}) = \text{pr}(Y^{\text{obs}} \mid \theta) = \int \text{pr}(Y^{\text{obs}}, Y^{\text{mis}} \mid \theta) dY^{\text{mis}}$$

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 - what else need to be imputed?

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- Idea: how to do inference with a single imputation
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- EM algorithm: impute the sufficient statistics

E-steps and M-steps

- Data: observed $\mathbf{Y}^{\text{obs}}$, missing $\mathbf{Y}^{\text{mis}}$; parameter: θ
- Complete data likelihood: $L(\theta; \mathbf{Y}^{\text{obs}}, \mathbf{Y}^{\text{mis}})$

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- E-step (given the current estimates of $\theta^{(t)}$)

$$Q(\theta \mid \theta^{(t)}) = \int \log L(\theta; \mathbf{Y}^{\text{obs}}, \mathbf{Y}^{\text{mis}} = y) \text{pr} \left\{ \mathbf{Y}^{\text{mis}} = y \mid \mathbf{Y}^{\text{obs}}, \theta^{(t)} \right\} dy$$

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$$\theta^{(t+1)} = \underset{\theta}{\operatorname{argmax}} Q(\theta \mid \theta^{(t)})$$

- Iterate until convergence

Example: bivariate normal

- Data: bivariate normal (Y_{i1}, Y_{i2}) with missing data in Y_{i1}

$$\begin{aligned}\log L(\theta \mid \mathbf{Y}_1, \mathbf{Y}_2) = & -\frac{n}{2} \{ \log(\sigma_1^2) + \log(\sigma_2^2) + \log(1 - \rho^2) \} \\ & + \sum_{i=1}^n \frac{1}{2(1 - \rho^2)} \left\{ \frac{(Y_{i1} - \mu_1)^2}{\sigma_1^2} - \frac{2\rho(Y_{i1} - \mu_1)(Y_{i2} - \mu_2)}{\sigma_1\sigma_2} + \frac{(Y_{i2} - \mu_2)^2}{\sigma_2^2} \right\}\end{aligned}$$

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- E-step: expectation of $\log L(\theta \mid \mathbf{Y}_1, \mathbf{Y}_2)$ given the observed data under $\theta = \theta^{(t)}$

$$\begin{aligned}& Q(\theta \mid \theta^{(t)}) \\ = & -\frac{n}{2} \{ \log(\sigma_1^2) + \log(\sigma_2^2) + \log(1 - \rho^2) \} \\ & + \frac{1}{2(1 - \rho^2)} \sum_{R_{i1}=1} \left\{ \frac{(Y_{i1} - \mu_1)^2}{\sigma_1^2} - \frac{2\rho(Y_{i1} - \mu_1)(Y_{i2} - \mu_2)}{\sigma_1\sigma_2} + \frac{(Y_{i2} - \mu_2)^2}{\sigma_2^2} \right\} \\ & + \sum_{R_{i1}=0} \left\{ \frac{\mathbb{E}\{(Y_{i1} - \mu_1)^2 \mid Y_{i2}\}}{2(1 - \rho^2)\sigma_1^2} - \frac{\rho\mathbb{E}\{(Y_{i1} - \mu_1)(Y_{i2} - \mu_2) \mid Y_{i2}\}}{(1 - \rho^2)\sigma_1\sigma_2} + \frac{(Y_{i2} - \mu_2)^2}{2(1 - \rho^2)\sigma_2^2} \right\}\end{aligned}$$

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- E-step:

- for units with $R_{i1} = 0$, we need to compute $\mathbb{E}(Y_{i1} \mid Y_{i2})$, $\mathbb{E}(Y_{i1}^2 \mid Y_{i2})$ and $\mathbb{E}(Y_{i1} Y_{i2} \mid Y_{i2})$

$$\mathbb{E}(Y_{i1} \mid Y_{i2}) = \mu_1^{(t)} + \rho^{(t)}(Y_{i2} - \mu_2^{(t)}), \quad \mathbb{E}(Y_{i1}^2 \mid Y_{i2}) = (1 - \rho^{(t)2})\sigma_1^{2(t)}$$

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- M-step:

$$\mu_1^{(t+1)} = \sum_{R_{i1}=1} Y_{i1} + \sum_{R_{i1}=0} \mathbb{E}(Y_{i1} | Y_{i2})$$

$$\sigma_1^{2(t+1)} = \frac{1}{n} \left\{ \sum_{R_{i1}=1} Y_{i1}^2 + \sum_{R_{i1}=0} \mathbb{E}(Y_{i1}^2 | Y_{i2}) - n\mu_1^{(t+1)2} \right\}$$

$$\rho^{(t+1)} = \frac{\sum_{R_{i1}=1} Y_{i1} Y_{i2} + \sum_{R_{i1}=0} \mathbb{E}(Y_{i1} | Y_{i2}) Y_{i2} - n\mu_1^{(t+1)} \mu_2^{(t+1)}}{n\sigma_1^{(t)} \sigma_2^{(t)}}$$

Example: bivariate normal

- To calculate $(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, we need the following sufficient statistics

$$\sum_{i=1}^n Y_{i1}, \quad \sum_{i=1}^n Y_{i2}, \quad \sum_{i=1}^n Y_{i1}^2, \quad \sum_{i=1}^n Y_{i2}^2, \quad \sum_{i=1}^n Y_{i1} Y_{i2}$$

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- EM algorithm: impute the sufficient statistics conditional on the observed data then apply the MLE formulas
 - $\{\mathbb{E}(Y_{i1} \mid Y_{i2})\}^2 \neq \mathbb{E}(Y_{i1}^2 \mid Y_{i2})$

Imputation methods

- Likelihood inference requires specifying the full model $\text{pr}(O_i; \theta)$
- Imputation method without full model: impute $\mathbf{O}^{\text{mis}}$
 - only requires $\text{pr}(\mathbf{O}^{\text{mis}} \mid \mathbf{O}^{\text{text}}; \theta)$
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- Single imputation: ignore the uncertainty in the imputation step
- Multiple imputation (Rubin, 1977, 1978)

Multiple imputation: example

- Example: estimation of $Q(\mathbf{O}) = \mathbb{E}(Y_i)$ from $(R_i = 1, Y_i, \mathbf{X}_i)$ and $(R_i = 0, \mathbf{X}_i)$
- Target from a Bayesian perspective:
$$\text{pr}(Q \mid Y^{\text{obs}}) = \int \text{pr}(Q \mid Y^{\text{mis}}, Y^{\text{obs}}) \text{pr}(Y^{\text{mis}} \mid Y^{\text{obs}}) dY^{\text{mis}}$$

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- Step 1: impute Y^{mis} for units with $R_i = 0$
 - sample Y_i^{mis} based on $\text{pr}(Y_i^{\text{mis}} \mid Y^{\text{obs}}, \mathbf{X}) \rightsquigarrow$ a complete dataset $\mathbf{O}_1$ is created
 - repeat m times to obtain $\mathbf{O}_1, \dots, \mathbf{O}_m$

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- Step 3: repeated imputation inference for Q

$$\begin{aligned}\mathbb{E}(Q \mid Y_i^{\text{obs}}) &= \mathbb{E} \left\{ \mathbb{E}(Q \mid Y_i^{\text{mis}}, Y_i^{\text{obs}}) \mid Y_i^{\text{obs}} \right\} \approx \text{average of } \hat{Q}_j \\ \text{var}(Q \mid Y_i^{\text{obs}}) &= \mathbb{E} \left\{ \text{var}(Q \mid Y_i^{\text{mis}}, Y_i^{\text{obs}}) \mid Y_i^{\text{obs}} \right\} + \text{var} \left\{ \mathbb{E}(Q \mid Y_i^{\text{mis}}, Y_i^{\text{obs}}) \mid Y_i^{\text{obs}} \right\} \\ &\approx \text{average of } \hat{\sigma}_j^2 + (1 + m^{-1}) \times \text{variance of } \hat{Q}_j\end{aligned}$$

Multiple imputation: discussion

- The theory is derived from a Bayesian perspective, but also justified by a frequentist perspective
- The first step, drawing random samples from an imputation model, is the fundamental step of the MI
 - requires the knowledge of $\text{pr}(Y_i^{\text{mis}} \mid Y_i^{\text{obs}}, \mathbf{X}_i)$
 - incorrectly used in many applications
 - a practical solution: multiple imputation by chained equations
- If we know the full model, we can use likelihood method or Bayesian inference

Multiple imputation by chained equations (MICE) (Raghunathan et al., 2001; Van Buuren, 2007)

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- Step 1: impute the mean for every missing value. These imputations can be thought as “place holders”
- Step 2: the “place holder” mean imputation for one variable (say Y_1) is set back to missing
- Step 3: regress the observed Y_1 on the other variables

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- Step 3: regress the observed Y_1 on the other variables
- Step 4: the missing values of Y_1 are replaced with **predictions** from the regression model
- Step 5: repeat steps 2-4 for each variable that has missing data. The cycling through each of the variables constitutes one iteration.
- Step 6: repeat steps 2-4 for until convergence

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 - Y_{i1} is set to missing for units with $R_{i1} = 0$

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 - impute $Y_{i1} = \mathbb{E}(Y_{i1} \mid R_{i1} = 1)$ for units with $R_{i1} = 0$, and $Y_{i2} = \mathbb{E}(Y_{i2} \mid R_{i2} = 1)$ for units with $R_{i2} = 0$
 - Y_{i1} is set to missing for units with $R_{i1} = 0$
 - Regress Y_{i1} on (Y_{i2}, Y_{i3}) for units with $R_{i1} = 1$, impute Y_{i1} for units with $R_{i1} = 0$ using the predictions: $\beta_1 Y_{i2} + \beta_2 Y_{i3} + \epsilon_i$
 - Regress Y_{i2} on (Y_{i1}, Y_{i3}) for units with $R_{i2} = 1$, impute Y_{i2} for units with $R_{i2} = 0$ using the predictions

Example

- A dataset with three variables: income (Y_1), age (Y_2), gender (Y_3)
- Income and age are MAR
- MICE procedure
 - impute $Y_{i1} = \mathbb{E}(Y_{i1} \mid R_{i1} = 1)$ for units with $R_{i1} = 0$, and $Y_{i2} = \mathbb{E}(Y_{i2} \mid R_{i2} = 1)$ for units with $R_{i2} = 0$
 - Y_{i1} is set to missing for units with $R_{i1} = 0$
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 - Regress Y_{i2} on (Y_{i1}, Y_{i3}) for units with $R_{i2} = 1$, impute Y_{i2} for units with $R_{i2} = 0$ using the predictions
 - repeat the imputation step until convergence

MICE: discussion

- Idea: by the end of the cycles the distribution of the parameters governing the imputations (e.g. the coefficients in the regression models) converge in the sense of becoming stable $\rightsquigarrow$ order does not matter

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- Number of imputed datasets depend on the size of the data, missing proportion and computational source
- Different MICE software packages vary in their exact implementation of this algorithm
- No theoretical justification $\rightsquigarrow$ a naive Gibbs Sampler?

Summary: general methods for missing data problems

- Complete data analysis
 - only use information from $(R_i = 1, Y_i, \mathbf{X}_i)$
 - valid under MCAR
 - can be tested by $\text{pr}(R = 1 \mid \mathbf{X}_i) = \text{pr}(R_i = 1)$
 - may be used when missing proportion is low (rule of thumb $< 5\%$)

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- Imputation methods
 - single imputation using mean, median, etc. $\rightsquigarrow$ ignore the uncertainty in the imputed data
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- Model-based methods $\rightsquigarrow$ models for $\text{pr}(Y_i, R_i \mid \mathbf{X}_i)$
 - models for $\text{pr}(Y_i \mid \mathbf{X}_i)$ and $\text{pr}(R_i \mid Y_i, \mathbf{X}_i)$
 - pattern mixture model: models for $\text{pr}(Y_i \mid R_i, \mathbf{X}_i)$ and $\text{pr}(R_i \mid \mathbf{X}_i)$
 - may work under MNAR, but requires correctly specified model
 - under MAR, models simplify as $\text{pr}(Y_i \mid \mathbf{X}_i)$ and $\text{pr}(R_i \mid \mathbf{X}_i)$
 $\rightsquigarrow$ no need to consider $\text{pr}(R_i \mid \mathbf{X}_i)$

A missing data perspective of ACE estimation

- $ACE = \mathbb{E}\{Y_i(1)\} - \mathbb{E}\{Y_i(0)\}$

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- Estimation of $\mathbb{E}\{Y_i(1)\}$
 - observed data: $(Z_i = 1, Y_i(1), \mathbf{X}_i)$ and $(Z_i = 0, \mathbf{X}_i)$
 - missing data indicator: Z_i .
- Estimation of $\mathbb{E}\{Y_i(0)\}$
 - observed data: $(Z_i = 0, Y_i(0), \mathbf{X}_i)$ and $(Z_i = 1, \mathbf{X}_i)$
 - missing data indicator: $1 - Z_i$.

A missing data perspective of ACE estimation

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 - observed data: $(Z_i = 0, Y_i(0), \mathbf{X}_i)$ and $(Z_i = 1, \mathbf{X}_i)$
 - missing data indicator: $1 - Z_i$.
- We can consider a general missing data problem
 - observed data: $(R_i = 1, Y_i, \mathbf{X}_i)$ and $(R_i = 0, \mathbf{X}_i)$
 - target: estimating $\mathbb{E}(Y_i)$

Estimation under MAR

- MAR: $Y_i \perp\!\!\!\perp R_i \mid X_i$
 - complete case not valid: $\mathbb{E}(Y_i \mid R_i = 1) \neq \mathbb{E}(Y_i)$
 - valid after covariate adjustment: $\mathbb{E}(Y_i \mid X_i, R_i = 1) = \mathbb{E}(Y_i \mid X_i)$

- Imputation: impute Y_i for units with $R_i = 0$

$$\begin{aligned}\mathbb{E}(Y_i) &= \mathbb{E}(Y_i \mid R_i = 1)\text{pr}(R_i = 1) + \mathbb{E}(Y_i \mid R_i = 0)\text{pr}(R_i = 0) \\ &= \mathbb{E}(Y_i \mid R_i = 1)p_1 + p_0 \int \mathbb{E}(Y_i \mid X_i = x, R_i = 0)F_0(dx) \\ &= \mathbb{E}(Y_i \mid R_i = 1)p_1 + p_0 \int \mathbb{E}(Y_i \mid X_i = x, R_i = 1)F_0(dx)\end{aligned}$$

- Imputation method (predictive)
 - step 1: fit model of Y_i on X_i for units with $R_i = 1$, denoted by $\hat{m}(x)$
 - step 2: impute $\hat{Y}_i = \hat{m}(X_i)$ for units with $R_i = 0$
 - step 3: $\frac{1}{n} \sum_{i=1}^n \left\{ Y_i \mathbf{1}(R_i = 1) + \hat{Y}_i \mathbf{1}(R_i = 0) \right\}$

Projection versus prediction

- Another type of imputation method

$$\mathbb{E}(Y_i) = \int \mathbb{E}(Y_i \mid X_i = x) dF(x) = \int \mathbb{E}(Y_i \mid X_i = x, R_i = 1) F(dx)$$

- Imputation method (projective)

- step 1: fit model of Y_i on X_i for units with $R_i = 1$, denoted by $\hat{m}(x)$
 - step 2: impute $\hat{Y}_i = \hat{m}(X_i)$ for all units
 - step 3: $\frac{1}{n} \sum_{i=1}^n \left\{ \hat{Y}_i \mathbf{1}(R_i = 1) + \hat{Y}_i \mathbf{1}(R_i = 0) \right\}$
-
- Two imputation methods are equivalent if linear model is used for estimating $\hat{m}(x)$: $\sum_{i=1}^n Y_i \mathbf{1}(R_i = 1) = \sum_{i=1}^n \hat{m}(X_i) \mathbf{1}(R_i = 1)$

Weighting

- Horvitz-Thompson estimation: a unit is weighted inverse-proportionally to its probability of being observed

$$\mathbb{E}(Y_i) = \mathbb{E} \left\{ \frac{Y_i R_i}{\text{pr}(R_i = 1 \mid X_i)} \right\}$$

- Weighting method
 - step 1: fit model of R_i on X_i , denoted by $\hat{\pi}(x)$
 - step 2: $\frac{1}{n} \sum_{i=1}^n \frac{Y_i R_i}{\hat{\pi}(X_i)}$
- Hajek estimator: replace the denominator n with $\sum_{i=1}^n \frac{R_i}{\hat{\pi}(X_i)}$

Imputation + weighting

- Model assisted estimation, augmented inverse probability weighting

$$\begin{aligned}\mathbb{E}(Y_i) &= \mathbb{E}\{m(X_i)\} + \mathbb{E}\{Y_i - m(X_i)\} \\ &= \mathbb{E}\{m(X_i)\} + \mathbb{E}\left[\frac{R_i\{Y_i - m(X_i)\}}{\pi(X_i)}\right]\end{aligned}$$

- Estimation strategy
 - step 1: estimate $\hat{m}(x)$ and $\hat{\pi}(x)$
 - step 2:

$$\frac{1}{n} \sum_{i=1}^n \left[\hat{m}(X_i) + \frac{R_i\{Y_i - \hat{m}(X_i)\}}{\hat{\pi}(X_i)} \right]$$

- Double robustness: unbiased estimation if either the model for $m(x)$ or $\pi(x)$ is correct

Imputation methods in causal inference

- Imputation method (predictive): impute $Y_i(0)$ for units with $Z_i = 1$ and $Y_i(1)$ for units with $Z_i = 0$

- let (p_1, p_0) be the proportions of treated and control units
- $m_z(x) = \mathbb{E}\{Y_i(z) \mid X_i = x\} = \mathbb{E}\{Y_i \mid Z_i = z, X_i = x\}$
- $F_z(x)$ is the cdf of X for units with $Z_i = z$

$$\begin{aligned}\text{ACE} &= p_1 \mathbb{E}\{Y_i(1) \mid Z_i = 1\} + p_0 \mathbb{E}\{Y_i(1) \mid Z_i = 0\} \\ &\quad - p_1 \mathbb{E}\{Y_i(0) \mid Z_i = 1\} - p_0 \mathbb{E}\{Y_i(0) \mid Z_i = 0\} \\ &= p_1 \mathbb{E}\{Y_i^{\text{obs}} \mid Z_i = 1\} + p_0 \int m_1(x) F_0(dx) \\ &\quad - p_1 \int m_0(x) F_1(dx) - p_0 \mathbb{E}\{Y_i^{\text{obs}} \mid Z_i = 0\}\end{aligned}$$

- Estimation strategy

- step 1: estimate $\hat{m}_z(x)$
- step 2: impute $\hat{Y}_i(z) = \hat{m}_z(X_i)$ for units with $Z_i = 1 - z$
- step 3: the predictive estimator

$$\frac{1}{n} \left\{ \sum_{i=1}^n Z_i Y_i(1) + \sum_{i=1}^n (1 - Z_i) \hat{Y}_i(1) - \sum_{i=1}^n Z_i \hat{Y}_i(0) - \sum_{i=1}^n (1 - Z_i) Y_i(0) \right\}$$

Imputation methods in causal inference

- Imputation method (projective): impute $Y_i(1)$ and $Y_i(0)$ for all units
 - $F(x)$ is the cdf of X_i for all units

$$\text{ACE} = \int m_1(x)F(dx) - \int m_0(x)F(dx)$$

- Estimation strategy
 - step 1: estimate $\hat{m}_z(x)$
 - step 2: impute $\hat{Y}_i(z) = \hat{m}_z(X_i)$ for all units
 - step 3: the projective estimator

$$\frac{1}{n} \left\{ \sum_{i=1}^n \hat{Y}_i(1) - \sum_{i=1}^n \hat{Y}_i(0) \right\}$$

- Two imputation methods are equivalent if linear model is used for estimating $\hat{m}_z(x)$
- Beyond linear model imputation: LASSO regression (Bloniarz *et al.*, 2016), machine learning (Wager *et al.*, 2016)

Weighting methods in causal inference

- Propensity score weighting ($e(x) = \text{pr}(Z_i = 1 \mid X_i = x)$)

$$\text{ACE} = \mathbb{E} \left\{ \frac{Z_i Y_i}{e(X_i)} \right\} - \mathbb{E} \left\{ \frac{(1 - Z_i) Y_i}{1 - e(X_i)} \right\}$$

- Estimation strategy

- step 1: estimate the propensity score $\hat{e}(x)$
- step 2: $\frac{1}{n} \left\{ \sum_{i=1}^n \frac{Z_i Y_i}{\hat{e}(X_i)} - \sum_{i=1}^n \frac{(1 - Z_i) Y_i}{1 - \hat{e}(X_i)} \right\}$

- Hajek estimator: replace n with $\sum_{i=1}^n \frac{Z_i}{\hat{e}(X_i)}$ or $\sum_{i=1}^n \frac{1 - Z_i}{1 - \hat{e}(X_i)}$

Imputation + Weighting in causal inference

- Model assisted propensity score weighting

$$\begin{aligned}\mathbb{E}\{Y_i(1)\} &= \mathbb{E}\{m_1(X_i)\} + \mathbb{E}\left[\frac{Z_i\{Y_i - m_1(X_i)\}}{e(X_i)}\right] \\ \mathbb{E}\{Y_i(0)\} &= \mathbb{E}\{m_0(X_i)\} + \mathbb{E}\left[\frac{(1 - Z_i)\{Y_i - m_0(X_i)\}}{1 - e(X_i)}\right]\end{aligned}$$

- Doubly robust estimator: projective estimator augmented by IPW
 - step 1: estimate $\hat{m}_z(x)$ and $\hat{e}(x)$
 - step 2:

$$\frac{1}{n} \sum_{i=1}^n \left[\hat{m}_1(X_i) + \frac{Z_i\{Y_i - \hat{m}_1(X_i)\}}{\hat{\pi}(X_i)} - \hat{m}_0(X_i) - \frac{(1 - Z_i)\{Y_i - \hat{m}_0(X_i)\}}{1 - \hat{\pi}(X_i)} \right]$$

Summary

- Treatment assignment mechanism is a special case of missing data mechanism
 - Randomized experiments $\rightsquigarrow$ MCAR
 - Ignorable treatment assignment $\rightsquigarrow$ MAR
 - Non-ignorable treatment assignment $\rightsquigarrow$ MNAR
- Imputation method
 - single imputation underestimates the variance
 - a principal method: multiple imputation
 - EM algorithm: impute the sufficient statistics
- The inferential framework of causal inference and missing data analysis are closely related, with common roots in survey sampling: imputation and weighting
- The concept of DR was first proposed in the context of missing data and was later formally extended to causal inference

Suggested readings

- Missing data mechanisms
 - Rubin 1976. “Inference and missing data”
 - Mealli and Rubin 2015. “Clarifying missing at random and related definitions, and implications when coupled with exchangeability”
- Multiple imputation
 - Azur *et al.* 2011. “ Multiple imputation by chained equations: what is it and how does it work?”
- EM algorithm
 - Dempster *et al.* 1977. “Maximum likelihood from incomplete data via the EM Algorithm”
- Missing data and causal inference
 - Ding and Li 2018. “Causal inference: A missing data perspective”